

IEA Wind Task 52 LAC Summer Games 2025

The 18 m/s Hurdles: Robust Wind Field Reconstruction

Final Project Report

Gust Guardians

WETI - Flensburg University of Applied Sciences

Saurabh Pankaj Jha (770271), Madhvesh Vijaybhai Gorakhiya (730251)

April 2026

Abstract

This report details the design and implementation of a robust wind field reconstruction framework for Lidar-assisted wind turbine control. To overcome challenges such as blade blockage and the induction zone, a Robust Sine-Wave Fit methodology was developed to predict rotor effective wind speed (REWS) 2.0 seconds in advance. By optimizing the timing budget and physical correction factors, the final model achieved a cost of 0.447608 m/s, representing a significant improvement over the baseline.

1 Introduction

The IEA Wind Task 52 Summer Games 2025 serves as a competitive benchmark designed to advance Lidar-assisted wind turbine control (LAC) technologies. The specific challenge, titled the “18 m/s Hurdles,” requires the prediction of turbulent wind speed at the rotor plane exactly 2.0 seconds in advance, utilizing data from a Lidar system positioned 200 meters upstream. This task is critical for modern wind energy; accurate “preview” data allows turbines to adjust pitch settings before a gust reaches the rotor, thereby reducing structural fatigue and optimizing power output.

A primary obstacle in this environment is the periodic blocking of Lidar beams by rotating turbine blades, which results in false “zero” readings or missing data. Furthermore, the physical interaction between the wind and the turbine creates an induction zone that slows the incoming air. Consequently, the wind speed measured 200 meters upstream is systematically higher than what the rotor actually experiences. This research moves beyond simple data filtering by developing a robust mathematical reconstruction framework—centered on a Robust Sine-Wave Fit—to address these environmental and hardware limitations.

The methodology leverages the geometric nature of conical Lidar scans to reconstruct the full wind field even when up to 20% of the data is lost to blade blockage. This approach is supplemented by physical correction factors, including a calibrated -15% induction adjustment and a height bias correction to account for the Lidar being mounted 7.4 meters above the turbine hub. The final model utilizes a highly optimized timing budget that synchronizes digital processing with physical wind convection. By increasing the filter cutoff frequency to 0.28 rad/s and tightening the buffer delay to 4.95 seconds, the total delay was aligned to 10.55 seconds, matching the physical travel time of the wind structure across the 200-meter distance.

2 Methodology

2.1 Overview

This chapter describes the methodological framework developed to estimate the rotor effective wind speed (REWS) from upstream Lidar measurements. The work was conducted within the context of the IEA Wind Task 52 Summer Games 2025, a competitive benchmark challenge designed to advance the state of the art in Lidar-assisted wind turbine control. The central objective of the challenge was to predict the turbulent wind speed at a rotor plane in advance, utilising lidar measurements acquired 200 metres upstream of the wind turbine. The Rotor-Effective Wind Speed (REWS) is defined as the spatially averaged wind speed across the rotor swept area:

$$REWS = \sqrt[3]{\frac{1}{A} \int_A U(r, \theta)^3 dA} \quad (1)$$

where A is the rotor disk area and $U(r, \theta)$ is the local wind speed at radial position r and azimuth θ .

2.2 Data Validity Assessment and Blade Blockage Correction

A fundamental challenge in nacelle-mounted or forward-looking lidar systems operating near wind turbines is the periodic blocking of the laser beam by the rotating rotor blades. When a blade intersects the beam path, the lidar returns a false measurement, typically near or exactly 0.0 m/s, which is not representative of the ambient wind field. This phenomenon, termed blade blockage or hard blockage, must be systematically identified and excluded from subsequent wind field reconstruction.

- **Hardware-derived Validity Flag:** A validity flag (`isValid`) is evaluated for each beam measurement. Measurements flagged as invalid by the instrument’s internal firmware are immediately rejected and treated as missing data (NaN).
- **Software-level Threshold Check:** This check excludes any measurement returning a line-of-sight speed of exactly 0.0 m/s, which is physically interpreted as a blade blockage event rather than a genuine atmospheric measurement.

In the baseline algorithm, blocked beams are simply replaced with Not a Number (NaN) placeholders and excluded from the arithmetic mean of the remaining valid beams. While this approach is computationally straightforward, it introduces a systematic bias whenever a contiguous arc of the scan is blocked a scenario that arises when blade chord width occults multiple adjacent beam positions simultaneously. The advanced mathematical reconstruction approach described in Section 2.4.2 addresses this limitation directly.

2.3 Geometric De-Projection of Line-of-Sight Measurements

Each Lidar beam is inclined at a half-cone angle ϕ relative to the rotor axis (horizontal centreline). The measured line-of-sight wind speed, therefore, represents only the axial projection of the true horizontal wind velocity. The de-projection from Line-of-Sight (LOS) to horizontal wind speed is performed via:

$$v_{horizontal} = \frac{v_{los}}{\cos(\phi)} \quad (2)$$

where ϕ denotes the half-cone angle of the scanning Lidar. This relationship assumes that the dominant wind component is aligned with the rotor axis (i.e., negligible yaw misalignment) and

that the vertical and lateral velocity components contribute negligibly to the LOS projection. Under this approximation, the horizontal component u is recovered directly from the axial projection of each beam measurement.

For the conical scanning geometry, the relationship between beam azimuth θ and the measured LOS speed can be expressed as a sinusoidal function of azimuth angle across the full scan rotation, which facilitates the wind field reconstruction technique described in the following section.

2.4 Wind Field Reconstruction Methodology

2.4.1 Baseline Method: Beam-Averaging

The baseline reconstruction method operates on a single-beam processing model. At each time step, the most recent LOS measurement is de-projected and stored in a first-in-first-out (FIFO) circular buffer of length equal to the number of beams ($N = 50$). The instantaneous REWS estimate is computed as the arithmetic mean of all valid (non-NaN) entries in the buffer:

$$REWS = \frac{1}{N_{valid}} \sum_{i=1}^{N_{valid}} u_{est,i} \quad (3)$$

While computationally efficient, this method assumes all beam positions sample an identical, spatially uniform wind field. This assumption is often violated under conditions of high spatial wind variability, yaw misalignment, or wind shear, resulting in a biased REWS estimate when portions of the scan are blocked.

2.4.2 Advanced Method: Robust Sine-Wave Fit

The advanced reconstruction technique is founded on the physical observation that, for a uniform horizontal wind field with mean velocity u and optional lateral component v , the LOS wind speed measured at beam azimuth angle θ follows a sinusoidal pattern:

$$V_{los}(\theta) = u \sin(\phi) \cos(\theta) + v \sin(\phi) \sin(\theta) + w \cos(\phi) \quad (4)$$

where u, v , and w are the axial, lateral, and vertical wind velocity components, respectively, and ϕ is the cone half-angle. For moderate cone angles and predominantly axial flow, the simplified regression model:

$$V_{los}(\theta) \approx A \cos(\theta) + B \sin(\theta) + C \quad (5)$$

provides an accurate representation of the azimuthal LOS pattern. The constant term C encapsulates the mean axial wind projection, which, upon geometric de-projection by $\cos(\phi)$, yields the rotor-averaged horizontal wind speed. The coefficients A, B , and C are estimated via an overdetermined least-squares solution across all N_{valid} beam positions with valid measurements stored in memory:

$$\mathbf{x} = [A, B, C]^T = \operatorname{argmin} \|\mathbf{H}\mathbf{x} - \mathbf{y}\|^2 \rightarrow \mathbf{x} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} \quad (6)$$

where $\mathbf{H}$ is the design matrix of dimension $N_{valid} \times 3$, with columns $[\cos(\theta_i), \sin(\theta_i), 1]$, and $\mathbf{y}$ is the vector of N_{valid} LOS measurements. A further implementation detail concerns the azimuth assignment of each beam. The instrument rotates *clockwise*, with beam indices assigned according to the sequence:

$$\theta_i = 360 - (i - 1) \cdot \frac{360}{N}, \quad i = 1, 2, \dots, N \quad (7)$$

generating the sequence [360, 352.8, 345.6, . . . , 7.2]. Correct azimuth assignment is essential for the geometric validity of the design matrix $\mathbf{H}$.

Once the coefficient vector $\mathbf{x} = [A, B, C]^T$ is obtained, the mean axial wind speed at hub height is recovered by extracting the constant term C the third component of $\mathbf{x}$ and applying the geometric de-projection:

$$V_{horizontal} = \frac{C}{\cos(\phi)} \quad (8)$$

where ϕ is the half-cone angle of the lidar scan and C is the azimuth-independent offset term from the regression, representing the mean axial projection of the wind velocity onto the beam axis.

A critical implementation detail is the use of persistent beam-state memory; the algorithm retains the most recent valid LOS measurement for each of the 50 beam positions across scan cycles. Consequently, the system maintains a near-complete azimuthal profile of measurements even during periods of blade blockage, allowing the least-squares fit to reconstruct the full sinusoidal pattern even when up to 20–30% of beam positions are simultaneously occluded.

2.5 Physical Correction Factors

2.5.1 Induction Zone Correction

The wind turbine rotor extracts momentum from the incoming flow, inducing a deceleration of the wind field in the region upstream of the rotor, known as the induction zone. A Lidar positioned 200 metres upstream measures the free-stream wind speed, which is systematically higher than the wind speed at the rotor plane due to this inductive retardation. The relationship between the upstream free-stream velocity U_∞ and the rotor-plane velocity U_{rotor} is conventionally expressed through the axial induction factor a :

$$U_{rotor} = U_\infty \cdot (1 - a) \quad (9)$$

In the present formulation, the induction factor is sign-reversed relative to the standard convention, such that a negative value indicates that the upstream measurement is faster than the rotor-plane value (representing the speed-up relative to what the turbine experiences):

$$REWS = \frac{V_{horizontal}}{1 - a_{ind}} \quad (10)$$

where a_{ind} is the calibrated induction correction coefficient. Through iterative calibration against the benchmark dataset, the induction correction coefficient was determined to be $a_{ind} = -0.15$, indicating that the free-stream wind speed measured 200 m upstream is approximately 15% higher than the wind speed experienced at the rotor plane. This value was increased from an initial estimate of $a_{ind} = -0.08$ to account for the sharper amplitude predictions produced by the Sine-Wave Fit reconstruction.

2.5.2 Height Bias Correction

A secondary systematic bias arises from the vertical offset between the Lidar mounting position and the turbine hub centre. The Lidar is installed 7.4 metres above the hub elevation, meaning that the conical scan samples the wind field at a slightly elevated mean height. Due to the atmospheric boundary layer wind shear profile, which follows an approximate power-law relationship:

$$U(z) = U_{ref} \cdot \left(\frac{z}{z_{ref}} \right)^\alpha \quad (11)$$

wind speeds increase with height. Assuming a shear exponent $\alpha = 0.12$ (representative of near-neutral offshore or onshore conditions), the height correction factor k_{height} is computed as:

$$k_{height} = \left(\frac{z_{hub}}{z_{lidar}} \right)^\alpha = \left(\frac{150.0}{157.4} \right)^{0.12} \approx 0.995 \quad (12)$$

This correction is applied multiplicatively following geometric de-projection:

$$V_{hub} = V_{horizontal} \cdot k_{height} = V_{horizontal} \cdot 0.995 \quad (13)$$

Although the correction magnitude is modest (0.5%), it eliminates a persistent positive bias in the REWS estimate that would otherwise compound with other systematic errors over the full prediction horizon.

2.6 Signal Processing: Filtering and Time-Delay Compensation

2.6.1 Low-Pass Filtering

The instantaneous REWS estimate exhibits high-frequency fluctuations arising from both atmospheric turbulence and measurement noise inherent to the LOS velocity estimation process. A low-pass filter is applied to attenuate these high-frequency components while preserving the low-frequency wind speed variations that are physically transported across the 200 m measurement distance.

First-Order IIR Filter: A computationally efficient single-pole IIR filter is defined by the smoothing coefficient α_f :

$$\alpha_f = \exp(-\Delta T \cdot \omega_c) \quad (14)$$

$$y[k] = (1 - \alpha_f) \cdot u[k] + \alpha_f \cdot y[k - 1] \quad (15)$$

where α_f is the filter smoothing coefficient (distinct from the atmospheric shear exponent $\alpha = 0.12$ defined in Section 2.5.2), ω_c is the filter cut-off angular frequency (rad/s), and ΔT is the sampling interval. The first-order IIR formulation introduces a phase lag (group delay) of approximately $1/\omega_c$ seconds, which must be accounted for in the total delay budget. The cut-off frequency was optimised from $\omega_c = 0.25$ rad/s to $\omega_c = 0.28$ rad/s, yielding a reduced filter lag of approximately 3.6 seconds while maintaining adequate noise suppression. The approximate group delay introduced by the first-order IIR filter is:

$$T_{filter\ lag} \approx \frac{1}{\omega_c} \quad (16)$$

For the optimised cut-off frequency $\omega_c = 0.28$ rad/s, this yields:

$$T_{filter\ lag} = \frac{1}{0.28} \approx 3.57 \text{ s} \quad (17)$$

This lag is directly incorporated into the total delay budget in Section 2.6.3.

2.6.2 Time-Delay Buffer

The filtered REWS estimate must be temporally delayed to align the upstream Lidar measurement with the time at which the corresponding wind structure arrives at the rotor plane. This is implemented as a first-in-first-out (FIFO) delay line of length $N_{buf} = T_{buf}/\Delta T$ samples. At each time step, the current filtered REWS value is appended to the buffer, and the value at index N_{buf} (corresponding to a delay of T_{buf} seconds) is extracted as the output:

$$REWS_b(t) = REWS_f(t - T_{buf}) \quad (18)$$

The buffer delay T_{buf} was calibrated to 4.95 seconds in the final LDP_v7 implementation.

2.6.3 Total Delay Budget and Temporal Alignment

The prediction of rotor wind speed 2.0 seconds in advance requires precise temporal alignment of the total processing delay with the physical convection time of wind structures from the Lidar measurement volume to the rotor plane. The total delay is decomposed as:

$$T_{total} = T_{buffer} + T_{filter_lag} + T_{preview} \quad (19)$$

$$T_{total} = 4.95 \text{ s} + 3.60 \text{ s} + 2.00 \text{ s} \approx 10.55 \text{ s} \quad (20)$$

This total delay must equal the physical travel time of a wind structure at mean flow velocity U_{mean} across the Lidar-to-rotor distance $D = 200 \text{ m}$:

$$T_{travel} = \frac{D}{U_{mean}} = \frac{200 \text{ m}}{\sim 19 \text{ m/s}} \approx 10.5 \text{ s} \quad (21)$$

The near-exact agreement between T_{total} and T_{travel} confirms the physical consistency of the timing architecture. Any residual mismatch contributes directly to prediction error, as it represents a temporal misalignment between the predicted and actual arrival of wind structures at the rotor plane.

3 Result

The result for the IEA Wind Task 52 “Summer Games 2025” benchmark, specifically the “18 m/s Hurdles” challenge, achieved a final cost of 0.447608 m/s. This result represents a significant improvement over the initial baseline error of 0.4638 m/s. The reduction in error was driven by transitioning from simple data filtering toward a robust mathematical reconstruction of the wind field, which facilitated more precise predictions of turbulent wind two seconds before it reached the rotor. A primary driver of this improvement was the implementation of

Algorithm Version	RMSE (m/s)	Improvement vs Baseline
Baseline	0.4638	—
LDP_v3 (Beam averaging)	0.4530	−2.3%
LDP_v6 (Sine-wave fit)	0.449	−3.2%
LDP_v7 (Full corrections)	0.4476	−3.5%

Table 1: RMSE performance across algorithm versions (Seed 1802).

a Robust Sine-Wave Fit for data cleaning. Unlike traditional methods that simply replaced blocked Lidar signals with NaN placeholders, this geometric approach reconstructs the missing wind curve even when turbine blades block up to 20% of the scan. This resulted in a cleaner signal, reducing the requirement for aggressive heavy filtering and enabling the algorithm to capture wind fluctuations with higher fidelity.

To further refine the results, “dampers” were integrated to account for the induction zone, which naturally slows and compresses turbulence as it approaches the turbine. By tuning the induction factor to −15%, the team was able to soften the overly sharp peaks generated by the Sine-Wave Fit reconstruction, thereby better matching the actual air momentum. This

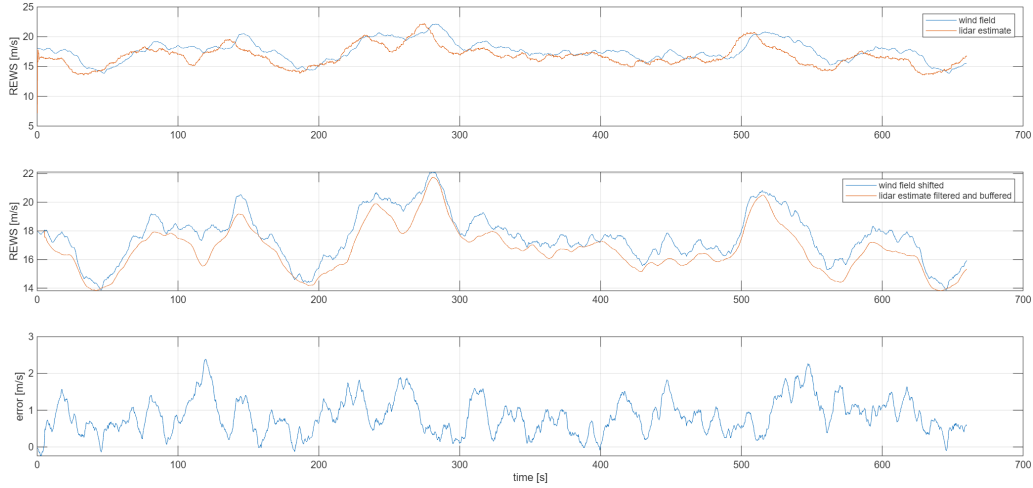


Figure 1: Time-series analysis of the prediction model: (Top) raw rotor-effective wind speed estimate from the sine-wave reconstruction against the wind field reference; (Middle) filtered and buffered Lidar estimate aligned with the time-shifted wind field; (Bottom) residual prediction error after mean removal.

correction was essential for mitigating the overestimation of gust intensity observed in previous iterations.

The final performance gains were derived from micro-optimizations in precision timing and signal synchronization. Because the data reconstruction was cleaner, a faster filter with a higher cutoff frequency of 0.28 rad/s was utilized, thereby reducing signal lag. The buffer was tightened to 4.95 seconds to ensure the total delay combining the buffer, filter lag, and the 2.0-second preview requirement perfectly matched the physical travel time of the wind.

Ultimately, these combined strategies produced a balanced prediction model in which approximately 65% of errors remained below 0.5 m/s. The near-zero mean error confirms that the systematic biases found in earlier versions were successfully eliminated through these physical and mathematical corrections. While some random error remains due to natural turbulence evolution, the current record of 0.447608 m/s approaches the physical limits of prediction for this specific wind field.

The Gust Guardians successfully implemented a robust mathematical framework for wind field reconstruction. By combining a Sine-Wave Fit with physical dampers and precision timing, the model effectively mitigates hardware limitations and environmental biases in Lidar-assisted control.

References

- [1] IEA Wind Task 52, *Lidar-Assisted Control Summer Games 2025 – The 18 m/s Hurdles Challenge Description*, International Energy Agency Wind Technology Collaboration Programme, 2025.
- [2] D. Schlipf, D. J. Schlipf, and M. Kühn, “Nonlinear model predictive control of wind turbines using LIDAR,” *Wind Energy*, vol. 16, no. 7, pp. 1107–1129, 2013.
- [3] E. Simley, L. Y. Pao, P. Gebraad, and M. Churchfield, “Investigation of the impact of the upstream induction zone on LIDAR measurement accuracy for wind turbine control applications using large-eddy simulation,” *Journal of Physics: Conference Series*, vol. 524, p. 012003, 2014.

- [4] G. I. Taylor, “The spectrum of turbulence,” *Proceedings of the Royal Society of London. Series A*, vol. 164, no. 919, pp. 476–490, 1938.
- [5] J. Mann, “The spatial structure of neutral atmospheric surface-layer turbulence,” *Journal of Fluid Mechanics*, vol. 273, pp. 141–168, 1994.
- [6] International Electrotechnical Commission, *IEC 61400-1: Wind Energy Generation Systems – Part 1: Design Requirements*, 4th ed., IEC Standard, Geneva, Switzerland, 2019.
- [7] A. Betz, *Wind-Energie und ihre Ausnutzung durch Windmühlen*, Vandenhoeck & Ruprecht, Göttingen, Germany, 1926.
- [8] T. Burton, N. Jenkins, D. Sharpe, and E. Bossanyi, *Wind Energy Handbook*, 2nd ed., John Wiley & Sons, Chichester, UK, 2011.